

HDOC DAY – 28

PROGRAM 1

Aim: To write a simple C program that inputs positive rows and columns, creates a 2D array, inputs elements column by column, and prints them in row order using one loop.

Algorithm:

1. Start the program.
2. Ask the user for rows and columns. Keep asking if the input is zero or negative.
3. Declare a 2D array using the input rows and columns.
4. Input elements column by column using nested loops (outer loop for columns, inner loop for rows).
5. Print all elements in row-major order using a single for loop.
6. End the program.

C CODE -

```
#include <stdio.h>
```

```
int main() {
```

```
    int rows, cols;
```

```
    // Step 1: Input rows and columns (must be greater than 0)
```

```
    printf("Enter number of rows: ");
```

```
    scanf("%d", &rows);
```

```
    printf("Enter number of columns: ");
```

```
scanf("%d", &cols);
```

```
while (rows <= 0 || cols <= 0) {  
    printf("Please enter positive values!\n");  
    printf("Enter number of rows: ");  
    scanf("%d", &rows);  
    printf("Enter number of columns: ");  
    scanf("%d", &cols);  
}
```

```
// Step 2: Declare 2D array
```

```
int a[rows][cols];
```

```
// Step 3: Input elements column by column
```

```
printf("\nEnter array elements (column by column):\n");
```

```
for (int c = 0; c < cols; c++) {  
    for (int r = 0; r < rows; r++) {  
        printf("Enter element at [%d][%d]: ", r, c);  
        scanf("%d", &a[r][c]);  
    }  
}
```

```
// Step 4: Print elements using a single loop
```

```
printf("\nElements in row-major order:\n");
```

```
for (int i = 0; i < rows * cols; i++) {  
    int r = i / cols;  
    int c = i % cols;  
    printf("%d ", a[r][c]);  
}  
printf("\n");  
  
return 0;  
}
```

CODE SCREENSHOT :

```
#include <stdio.h>

int main() {
    int rows, cols;

    // Step 1: Input rows and columns (must be greater than 0)
    printf("Enter number of rows: ");
    scanf("%d", &rows);
    printf("Enter number of columns: ");
    scanf("%d", &cols);

    while (rows <= 0 || cols <= 0) {
        printf("Please enter positive values!\n");
        printf("Enter number of rows: ");
        scanf("%d", &rows);
        printf("Enter number of columns: ");
        scanf("%d", &cols);
    }

    // Step 2: Declare 2D array
    int a[rows][cols];

    // Step 3: Input elements column by column
    printf("\nEnter array elements (column by column):\n");
    for (int c = 0; c < cols; c++) {
        for (int r = 0; r < rows; r++) {
            printf("Enter element at [%d][%d]: ", r, c);
            scanf("%d", &a[r][c]);
        }
    }

    // Step 4: Print elements using a single loop
    printf("\nElements in row-major order:\n");
    for (int i = 0; i < rows * cols; i++) {
        int r = i / cols;
        int c = i % cols;
        printf("%d ", a[r][c]);
    }
    printf("\n");

    return 0;
}
```

OUTPUT SCREENSHOT :

```

~ % nano 28q1.c
~ % gcc 28q1.c -o 28q1
~ % ./28q1

Enter number of rows: -1
Enter number of columns: 2
Please enter positive values!
Enter number of rows: 2
Enter number of columns: 3

Enter array elements (column by column):
Enter element at [0][0]: 10
Enter element at [1][0]: 40
Enter element at [0][1]: 20
Enter element at [1][1]: 40
Enter element at [0][2]: 30
Enter element at [1][2]: 60

Elements in row-major order:
10 20 30 40 40 60

~ % █

```

PROGRAM 2 :

Aim: To write a C program that inputs a 2D array, copies its elements into a 1D array, and displays both arrays.

Algorithm:

1. Start the program.
2. Read rows and columns from the user.
3. Declare a 2D array and a 1D array.
4. Input elements into the 2D array.
5. Copy all elements from the 2D array to the 1D array using a counter variable.

6. Print the 2D array and then print the 1D array.
7. End the program.

C CODE :

```
#include <stdio.h>
```

```
int main() {
```

```
    int rows, cols;
```

```
    // Step 1: Input matrix size
```

```
    printf("Enter number of rows: ");
```

```
    scanf("%d", &rows);
```

```
    printf("Enter number of columns: ");
```

```
    scanf("%d", &cols);
```

```
    int a[rows][cols];
```

```
    int b[rows * cols];
```

```
    // Step 2: Input elements of 2D array
```

```
    printf("\nEnter elements of 2D array:\n");
```

```
    for (int i = 0; i < rows; i++) {
```

```
        for (int j = 0; j < cols; j++) {
```

```
            printf("Element [%d][%d]: ", i, j);
```

```
            scanf("%d", &a[i][j]);
```

```
}  
}
```

```
// Step 3: Copy 2D array into 1D array
```

```
int k = 0;  
for (int i = 0; i < rows; i++) {  
    for (int j = 0; j < cols; j++) {  
        b[k] = a[i][j];  
        k++;  
    }  
}
```

```
// Step 4: Display 2D array
```

```
printf("\n2D Array:\n");  
for (int i = 0; i < rows; i++) {  
    for (int j = 0; j < cols; j++) {  
        printf("%d ", a[i][j]);  
    }  
    printf("\n");  
}
```

```
// Step 5: Display 1D array
```

```
printf("\n1D Array:\n");
```

```
for (int i = 0; i < rows * cols; i++) {  
    printf("%d ", b[i]);  
}  
printf("\n");  
  
return 0;  
}
```

Code screenshot :

```
#include <stdio.h>

int main() {
    int rows, cols;

    // Step 1: Input matrix size
    printf("Enter number of rows: ");
    scanf("%d", &rows);
    printf("Enter number of columns: ");
    scanf("%d", &cols);

    int a[rows][cols];
    int b[rows * cols];

    // Step 2: Input elements of 2D array
    printf("\nEnter elements of 2D array:\n");
    for (int i = 0; i < rows; i++) {
        for (int j = 0; j < cols; j++) {
            printf("Element [%d][%d]: ", i, j);
            scanf("%d", &a[i][j]);
        }
    }

    // Step 3: Copy 2D array into 1D array
    int k = 0;
    for (int i = 0; i < rows; i++) {
        for (int j = 0; j < cols; j++) {
            b[k] = a[i][j];
            k++;
        }
    }

    // Step 4: Display 2D array
    printf("\n2D Array:\n");
    for (int i = 0; i < rows; i++) {
        for (int j = 0; j < cols; j++) {
            printf("%d ", a[i][j]);
        }
        printf("\n");
    }

    // Step 5: Display 1D array
    printf("\n1D Array:\n");
    for (int i = 0; i < rows * cols; i++) {
        printf("%d ", b[i]);
    }
    printf("\n");

    return 0;
}
```

OUTPUT SCREENSHOT :

```
| % gcc 28q2.c -o 28q2
| % ./28q2
Enter number of rows: 3
Enter number of columns: 2

Enter elements of 2D array:
Element [0][0]: 5
Element [0][1]: 15
Element [1][0]: 25
Element [1][1]: 35
Element [2][0]: 45
Element [2][1]: 55

2D Array:
5 15
25 35
45 55

1D Array:
5 15 25 35 45 55
| %
```